Report 1: Attention Task Validation

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March 10, 2026

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1 Introduction

1.1 Domain Background

Selective attention is the cognitive ability to focus on relevant stimuli while filtering out distractors. It is fundamental to everyday functioning and is

impaired in conditions such as ADHD and following traumatic brain injury.

The standard laboratory method for measuring selective attention is the *visual search task*: a participant views a display of objects and must find a target as quickly and accurately as possible [2]. Feature Integration Theory predicts that increasing the number of targets increases attentional load, producing slower reaction times and lower accuracy.

Laboratory-based assessment, while valid, has practical limitations. It requires specialised hardware, trained staff, and in-person attendance. Gamified mobile assessments have been proposed as a scalable alternative, but must be validated against established lab measures before they can be trusted [1].

1.2 Study Objectives

This study aims to validate a Gamified Selective Attention Task by comparing performance outcomes against a standard laboratory measure. A 2 × 2 mixed factorial design was employed, with Target Load (Single vs. Multiple) as a between-subjects variable and Modality (Lab vs. Game) as a within-subjects variable.

The research addresses the following aims:

1. **RQ1 — Concurrent Validity:** Is there a positive correlation between performance on the game and the lab task?

2. **RQ2 — Load Effect:** Does the Multiple Target condition produce poorer performance than the Single Target condition?

3. **RQ3 — Modality Effect:** Does the game interface systematically affect performance compared to the lab interface?

4. **RQ4 — Level Effect:** Does performance change across progressive game levels?

1.3 Research Hypotheses

H1 — Concurrent Validity

H_0: There is no correlation between performance on the Game and Lab task ($r = 0$).

H_1: Performance on the Game task is positively correlated with performance on the Lab task ($r > 0$, one-tailed).

H2 — Load Effect

H_0: RT and accuracy do not differ between the Single Target and Multiple Target conditions ($\mu_{\text{single}} = \mu_{\text{multiple}}$).

H_1: The Multiple Target condition results in slower RTs and lower accuracies compared to the Single Target condition, reflecting increased attentional load ($\mu_{\text{multiple}} > \mu_{\text{single}}$ for RT; $\mu_{\text{multiple}} < \mu_{\text{single}}$ for accuracy, one-tailed).

H3 — Modality Effect

H_0: There is no systematic difference in performance between the Lab and Game interfaces ($\mu_{\text{lab}} = \mu_{\text{game}}$).

H_1: Performance differs systematically between the Lab and Game interfaces ($\mu_{\text{lab}} \neq \mu_{\text{game}}$, two-tailed).

H4 — Level Effect

H_0: Performance is stable across game levels, with no linear relationship between level and RT ($r = 0$).

H_1: RT changes systematically across game levels ($r \neq 0$, two-tailed).

H5 — Target Colour Effect

H_0: Target colour has no effect on RT or accuracy ($\mu_{\text{red}} = \mu_{\text{other}}$).

H_1: Red targets result in faster RTs and higher accuracy compared to other colours, consistent with Feature Integration Theory ($\mu_{\text{red}} < \mu_{\text{other}}$ for RT; $\mu_{\text{red}} > \mu_{\text{other}}$ for accuracy, one-tailed).

1.4 Participants

$N = 37$ participants were randomly assigned to one of two Target Load groups. To control for practice and fatigue effects, the order of modalities (Lab first or Game first) was counterbalanced within each group.

Group	Condition	n	IDs
Group A	Single Target	21	1–21
Group B	Multiple Target	16	22–37

Table 1: Participant allocation

2 Methods

2.1 Independent and Dependent Variables

The experiment used a 2 × 2 Mixed Factorial Design with the following variables:

1. **Independent Variables (IVs)**

- **Target Load:** (Between-Subjects) Participants were assigned to either *Single Target* (one object to find) or *Multiple Target* (several objects to find at once).
- **Modality:** (Within-Subjects) Every participant played the *Game* on a smartphone and did the *Lab* task on a computer.

2. **Dependent Variables (DVs)**

- **Reaction Time (RT):** The time in seconds it takes from when the target appears to when the participant first clicks or taps it.
- **Accuracy:** The proportion (0 to 1) of correct clicks or taps out of everything the participant did.

2.2 Relevant Data Columns

The analysis uses a single dataset combining computer data and smartphone data. The critical columns evaluated include:

- `participant`**:** An integer identifier (1–37) consistent across all modalities ensuring proper within-subjects matching.
- `target_load`**:** Derived from the experimental condition assigned to the participant (Single or Multiple targets).

- `modality`: The source of the data (Lab or Game).
- `trial_num`: Extracted from `trials.thisN` (Lab) or `Level` (Game), representing the zero-indexed trial counter or sequential game level, respectively.
- `RT`: The primary timing measure in seconds. For lab data, this represents latency from true stimulus onset to first response (`mouse.time[0]`). For game data, it is derived from `InitialResponseTime(ms)` / 1000.
- `accuracy`: The primary correctness measure as a proportion (0–1). In the lab, it takes the value of 1 if the target object appears in the first entry of `mouse.clicked_name`. In the game, it is derived from `SuccessRate(%)` / 100.
- `target_col` (Lab Data Only): The specified colour of the target object required to test target saliency effects (H5).

2.3 Reaction Time

Reaction time (RT) was computed as:

$$RT = mouse.time[0] - (square.started - trial.started)$$

For game files:

$$RT = \frac{InitialResponseTime(ms)}{1000}$$

2.4 Accuracy

- **Single target:** Correct if `target` appears in `mouse.clicked_name`.
- **Multiple targets:** Correct objects are `target_0` to `target_4`. Accuracy = correct selections divided by 5.

For game files:

$$Accuracy = \frac{SuccessRate(\%)}{100}$$

2.5 Data Cleaning

1. Remove non-trial rows where `trials.thisN` is missing.
2. Exclude RT < 0.1 s or > 30 s.
3. Clip accuracy to the range $[0, 1]$.

3 Results

3.1 Descriptive Statistics

			RT (s)					Accuracy		
Target Load	Modality	N	M	SD	Med	Min	Max	M	SD	Med
Multiple	Game	16	1.856	0.507	1.719	1.136	3.120	0.964	0.025	0.963
	Lab	16	1.505	0.268	1.428	1.186	2.023	0.951	0.135	0.993
Single	Game	21	3.104	0.625	2.923	2.258	4.683	0.983	0.028	1.000
	Lab	21	1.495	0.225	1.471	1.093	2.223	1.000	0.000	1.000

Table 2: Descriptive statistics

3.2 Figures

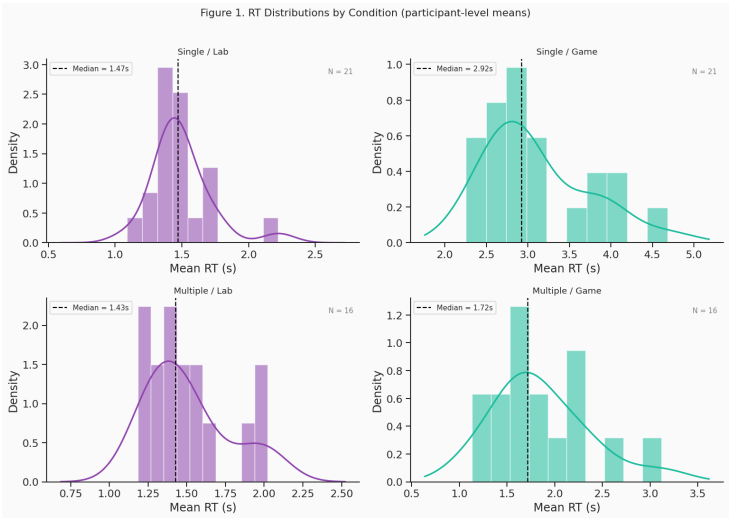

Figure 1: Reaction Time Distribution

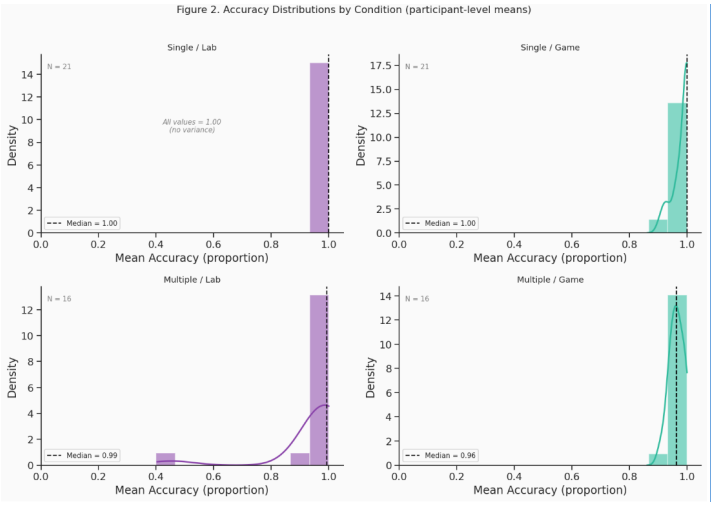

Figure 2: Accuracy Distribution

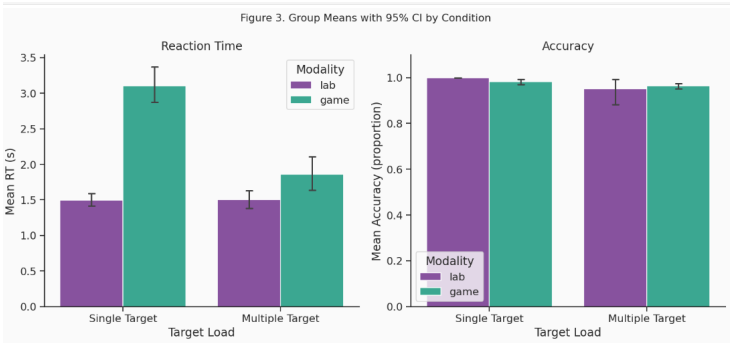

Figure 3: Group Distribution

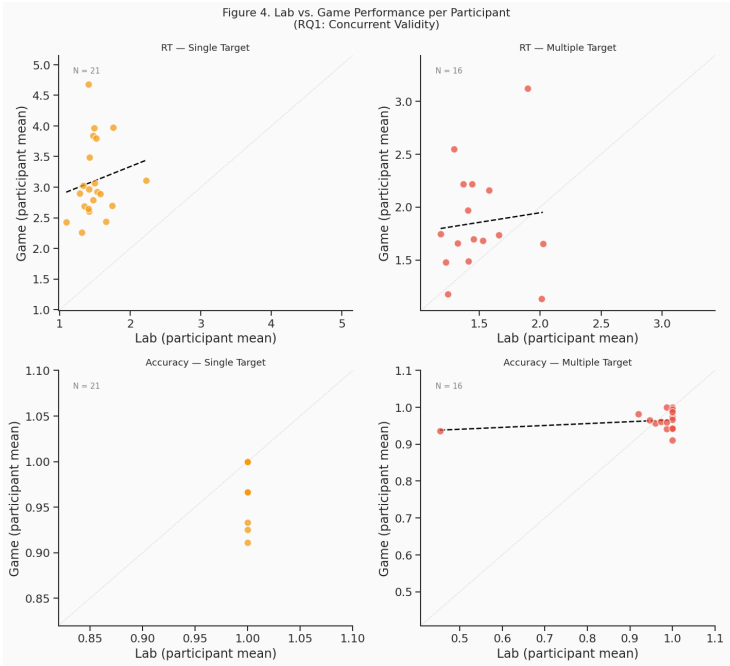

Figure 4: Lab vs Game Performance

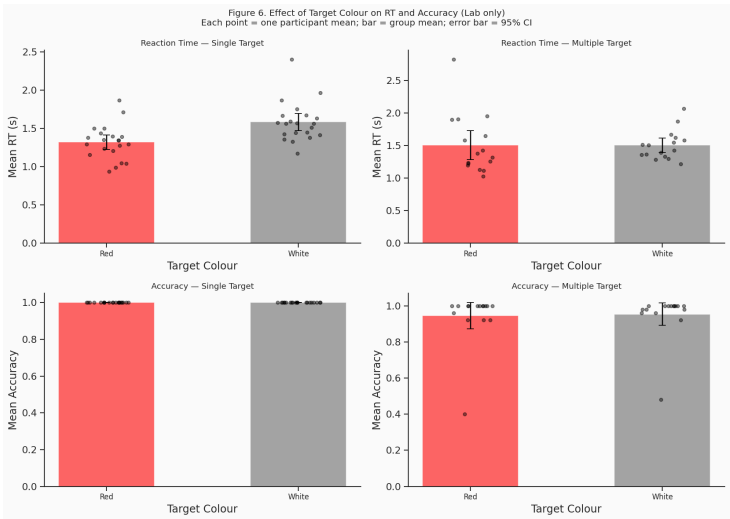

Figure 5: Effect of Target Colour

4 Future Work

Future work will include:

- A full 2×2 mixed ANOVA
- Non-parametric analyses where assumptions are violated
- Reliability analysis of both instruments

References

[1] Lumsden et al. (2016). Gamification of cognitive assessment and training. *JMIR Serious Games*.

[2] Wolfe (1994). Guided search 2.0. *Psychonomic Bulletin & Review*.